Ayush Agarwal

agarw534@purdue.edu | +1 408-375-8756 | linkedin.com/in/ayush-agarwal07 | github.com/Ayush-Agarwal07

Education

B.S. Computer Science — GPA: 3.8/4.0 *Expected May 2029*
Purdue University, College of Science West Lafayette, IN

- **Coursework:** Proof-based Linear Algebra, Discrete Mathematics, Data Structures & Algorithms, Systems Programming in C, Object-Oriented Programming in Java, Calculus I–III
- **Awards:** AIME Qualifier, AMC Distinction

Experience

Research Assistant — Prof. Tianyi Zhang *Jan 2026 – Present*
Purdue University West Lafayette, IN

- Conducting rigorous end-to-end benchmarking of LLM-based systems, systematically cataloging failure modes across diverse taxonomies and synthesizing findings into weekly research presentations for a team of 3 PhD students
- Developing and testing hypotheses about agent memory architecture through controlled perturbation experiments, isolating which signals are most predictive of performance improvement across diverse problem sets
- Synthesizing 30+ academic papers to identify research gaps, form original hypotheses, and make evidence-based architecture decisions — developing the habit of grounding conclusions in primary sources

Software Engineering Intern *Jan 2025 – Jun 2025*
MightyBot Palo Alto, CA

- Built and shipped RepoChat, an AI-powered analysis platform serving GPT-4o queries over GitHub repositories; owned the full product lifecycle from architecture through deployment and iterative improvement
- Engineered query decomposition layer with LangChain agents and Pydantic schemas that routes complex multi-file questions into file-scoped subtasks, improving answer relevance **30%+** over baseline as measured by an LLM-as-judge evaluation framework
- **Cut end-to-end latency 60%+** by diagnosing root causes of underperformance and implementing targeted fixes, demonstrating the value of rigorous analysis before jumping to solutions
- Built synthesis layer aggregating distributed outputs into coherent answers, improving comprehensiveness **25%+**; designed internal evaluation suite to continuously track and validate system performance

Software Engineering Intern *Dec 2023 – Feb 2024*
Awareye Palo Alto, CA

- Raised production CV model precision by **5%** through targeted data augmentation strategies and systematic hyperparameter search on an edge-deployed real-time object detection pipeline running on constrained hardware
- Shipped cross-platform mobile applications in React Native and Swift across multiple enterprise client deployment cycles; authored end-to-end regression and integration test suites validating UI and vision features against client acceptance criteria

Research Assistant — Prof. Koushik Sen *Oct 2022 – Jul 2023*
UC Berkeley Berkeley, CA

- Designed and maintained a Python data mining pipeline over 100+ GitHub repositories to extract, deduplicate, and cluster code snippets for a code idiom similarity model; research contributed to arxiv.org/abs/2312.15157

Projects

Project Lead, Boiler Quant *Jan 2026 – Present*
AgentQR — Python, LangChain, RAG, Backtesting

- Leading a 5-person team building a quantitative research system that autonomously surfaces non-obvious cross-sector trading signals — e.g., identifying cloud infrastructure capex as a leading indicator of semiconductor stock returns
- Designed adversarial debate framework where specialist agents challenge each other's investment theses before a synthesis agent reaches a conclusion — mirroring the buy-side research process; backtesting on 5 years of data yields Sharpe ∼0.75
- Grounded every agent-generated signal in primary financial sources — SEC filings, earnings transcripts, live market data — with retrieval pipelines ensuring all conclusions are traceable to verifiable, timestamped documents

Co-founder *Sep 2022 – Jun 2025*
Disadus — Next.js, React, Node.js, MongoDB

- Founded and scaled a learning management platform to 500+ active students across 8 schools; identified a market gap, built a solution, and grew it through direct relationships with school administrators
- Validated product improvements through 90-day post-launch cohort analysis, reducing navigation errors 25% and increasing session duration 15% — tracking outcomes with the same rigor applied to any investment thesis

Technical Skills

Languages: Python, JavaScript/TypeScript, C, C++, Java, Swift
Quant & Research: Pandas, NumPy, backtesting frameworks, financial data pipelines, RAG, LLM-as-judge evaluation
Tools: Git, Docker, FastAPI, PyTest, OpenAI API, Gemini API, TensorFlow, PyTorch, LangChain